

Factor Revealing LPs and Stable Matching with Ties and Incomplete Lists

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Abstract

Stable matching with ties and incomplete lists (SMTI) is one of the most prominent NP-hard problems in the domain of ordinal matching, where we seek to find a bipartite matching between a set of men and a set of women that is “stable” with respect to their preferences. When ties in preference lists are restricted to one side of the problem, Iwama et al. [10] devised a variant of the famous Gale-Shapley stable matching algorithm that breaks ties using edge weights from a linear programming (LP) relaxation of the problem, leading to an approximation ratio of $\frac{25}{17} \approx 1.4706$. We apply ideas from factor-revealing LPs to show, via computational proof involving the solution of massive LPs, that their analysis can be systematically improved to yield an approximation ratio of $19/13 \approx 1.4615$, improving the best currently-known ratio (obtained via different techniques in [16]) of $\frac{41}{28} \approx 1.4643$.

1 Introduction

Consider an instance I of a bipartite matching problem over a graph $G = (L \cup R, E)$, where L and R represent sets of men and women. Each man $m \in L$ and woman $w \in R$ submits a ranked preference list over potential partners in E . In the context of a matching $M \subseteq E$, edge $(m, w) \in E$ is a *blocking pair* if m is unmatched or prefers w to his partner in M , and similarly w is unmatched or prefers m to her partner in M . Informally, m and w would be unhappy if we imposed the matching M , since they would prefer to be matched with each-other. We say M is *stable* if no blocking pairs exist. Stable matchings and their variants have been a vibrant subfield of algorithmic game theory since the seminal work of Gale and Shapley [2] in the 1960s. In particular:

- Gale and Shapley proposed a simple algorithm that computes a stable matching in $O(|E|)$ time (i.e., linear time).
- If preference lists are strictly ordered (no ties) and *complete* ($E = L \times R$), then the Gale-Shapley (GS) algorithm finds a matching M of maximum possible size, with $|M| = \min(|L|, |R|)$. Otherwise, if preference lists are not complete, the resulting matching may include both unmatched men and women, but the unmatched men and women are the same in every stable matching, so all stable matchings have equal cardinality [1].
- If preference lists are complete but contain ties, the GS algorithm still computes a matching of size $\min(|L|, |R|)$. Here, stability is typically defined such that an edge $(m, w) \notin M$ is blocking if m is unmatched or *strictly* prefers w to his partner in M , and likewise for w .

Stable matching with both ties and incomplete lists (SMTI) is more challenging. Here, stable matchings may differ in cardinality, as shown in Figure 1, and the problem of computing a

maximum-cardinality stable matching is NP-hard [8, 13], even with ties restricted to one side of the problem (in preference lists of the women, as is the typical convention). Since one-sided ties and incomplete preference lists are anticipated in many ordinal matching problems in practice (e.g., [7]), the SMTI problem has received substantial attention recently, and is currently one of the most actively-studied problems in the ordinal matching research community. In this paper, we show a polynomial-time algorithm with the strongest approximation ratio for this problem to date, $19/13 \approx 1.4615$. Our approach is somewhat distinct in that we rely on the computational solution of a massive “factor-revealing” LP to systematically generalize and strengthen the analysis of Iwama et al. [10]. The application of factor-revealing LPs to online bipartite matching has recently appeared in [3, 4], but to the best of our knowledge, this work is the first involving factor-revealing LPs in the domain of stable matching.

Background. The GS algorithm produces a *maximal* stable matching, since any edge $(m, w) \in E$ between two unmatched individuals m and w would be blocking. It therefore gives a 2-approximate solution, owing to the well-known fact that a maximum matching can be at most twice the cardinality of any maximal matching. Using local search techniques, Iwama et al. [9] improved the factor to $\frac{15}{8}$ for the general case where ties are allowed on both sides. Király [11] introduced the idea of promoting unmatched men to higher levels of priority to break ties, achieving a ratio of $\frac{5}{3}$ for the general case and $\frac{3}{2}$ for the case where ties are restricted to one side. McDermid [14] improved the factor for the general case to $\frac{3}{2}$ by exploiting a classical graph theory result known as the Gallai-Edmonds decomposition, and Paluch [15] and Király [12] gave linear time algorithms obtaining the same ratio. Following this, Iwama, Miyazaki, and Yanagisawa [10] improved the approximation factor for the case with one-sided ties to $\frac{25}{17} \approx 1.4706$, by using the edge weights in a fractional linear programming (LP) relaxation of the problem to break ties in a natural way while running the GS algorithm. More recently, paper, Huang et al. [6] used different techniques to improve the ratio for the one-sided case to $\frac{22}{15} \approx 1.4667$; the analysis was tightened to $\frac{41}{28} \approx 1.4643$ by Radnai [16]. In terms of hardness results, the SMTI problem with two-sided ties cannot be approximated to a ratio better than $\frac{33}{29}$ unless $P = NP$, or $\frac{4}{3}$ assuming the unique games conjecture [19]. These results even apply when each individual has a single tie of length two. For the one-sided variant, we cannot approximate with a ratio better than $\frac{21}{19}$ unless $P = NP$ [19], or $\frac{5}{4}$ assuming the unique games conjecture [5].

2 Linear Programming Relaxation

The following LP relaxation of the one-sided SMTI problem was initially studied in [17] and [18]. The notation $m \succ_w m'$ indicates that m is preferred to m' by w .

$$LP(I) = \max \sum_{(m,w) \in E} x_{mw}$$

$$\text{subject to: } \sum_{w:(m,w) \in E} x_{mw} \leq 1 \quad \forall m \in L \quad (1)$$

$$\sum_{m:(m,w) \in E} x_{mw} \leq 1 \quad \forall w \in R \quad (2)$$

$$\sum_{w' \succ_m w} x_{mw'} + \sum_{m' \succeq_w m} x_{m'w} \geq 1 \quad \forall (m, w) \in E \quad (3)$$

$$x_{mw} \geq 0 \quad \forall (m, w) \in E$$

For each edge $(m, w) \in E$, the variable $x_{mw} \in [0, 1]$ indicates the extent to which m and w are matched. Constraints (1) and (2) are standard for matching problems, and ensure that each person can be matched to at most one other person. Constraint (3) enforces stability if

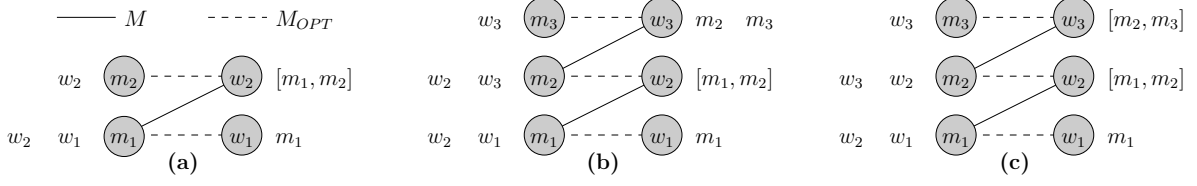


Figure 1: Simple instances admitting two stable matchings M and M_{OPT} of different cardinality: (a) an “augmenting 3-path”, and (b)-(c) “augmenting 5-paths”. Preference lists are shown alongside each man and woman, with ties in brackets.

x_{mw} ’s are restricted to integers, since it states that for every edge $(m, w) \in E$, at least one of m or w should be matched to a partner that is equally or more preferred than along this edge. Since the SMTI problem is exactly modeled by the formulation above with the added constraint that $x_{mw} \in \{0, 1\}$ for all $(m, w) \in E$, we have $LP(I) \geq |M_{OPT}(I)|$ for any instance I , where $M_{OPT}(I)$ is a maximum-cardinality stable matching for I .

Iwama et al. [10] show that the integrality gap of this LP relaxation is at least $1 + \frac{1}{e} \approx 1.3679$. Any analysis based on comparing to the objective of this LP relaxation therefore cannot lead to a stronger approximation guarantee.

3 The GS-LP Algorithm

Our algorithm is essentially the same as used by Iwama et al. [10], which is built from the standard GS algorithm by adding extra machinery to decide how to break ties. Since the GS algorithm is known to generate a stable matching irrespective of how ties are broken, we are therefore still guaranteed to find a stable matching. The standard GS algorithm is quite straightforward to describe: in each iteration, an arbitrary unassigned man proposes to the next woman on his preference list. Each woman tentatively holds on to the best proposal she has received to date, rejecting all other offers. When a man is rejected, he continues issuing proposals down his list. Each edge is proposed along at most once, leading to a linear running time.

3.1 A Prioritization Scheme for Tie-Breaking

As a preprocessing step, we solve the LP relaxation to obtain a solution x^* . Each man m is then assigned integer and fractional priority levels $P_i(m) \in \{0, 1, 2, \dots\}$ and $P_f(m) \in [0, 1]$, both initially zero. These priorities are used to resolve ties: if w is tentatively matched with m but receives a proposal from a man m' for which $m =_w m'$, she accepts if $P_i(m') > P_i(m)$, or if $P_i(m) = P_i(m')$ but $P_f(m') > P_f(m)$.

Let $w(m)$ denote the farthest woman down m ’s preference list up to whom he has proposed; initially, $w(m) = \emptyset$. The fractional priority for m is defined as $P_f(m) = \sum_{w \succeq_m w(m)} x_{mw}^*$. Every time m is rejected by $w(m)$, he returns to the start of his preference list and has another chance to propose to all the women up to $w(m)$ in sequence (he may be more successful this time, since his fractional priority now includes the contribution due to $x_{m, w(m)}^*$). If m runs off the end of his preference list (after being rejected by the final woman and then proposing one last time to everyone on the list), he increments the value of his integer priority $P_i(m)$, resets $w(m)$ to $\emptyset$ (so $P_f(m)$ resets to zero as well), and restarts the proposal process from the beginning of his list. Men whose priorities reach a specified threshold T cease to issue any further proposals, and the

algorithm terminates when every unmatched man m has $P_i(m) = T$. The algorithm of Iwama et al. uses $T = 3$, but we consider using higher thresholds; it turns out that $T = 4$ is sufficient for our analysis.

The entire algorithm clearly runs in polynomial time. Speedups to the GS-based part above may certainly be possible, although these may offer diminishing returns, since the solution of the LP relaxation will likely dominate the running time.

3.2 Integer Prioritization and Restricted Augmenting Paths

Many prior approximation algorithms for the SMTI problem are structured as above, using the GS algorithm along with some method of prioritization to break ties. The work of Paluch [15] and Kiraly [12] used essentially only the integer part of the prioritization scheme above (for which the LP solution is not needed), and leads to an approximation bound of $3/2$ due to the following observation on the resulting matching M . This holds in our case as well since we use the same integer prioritization machinery, and it takes precedence over any tie-breaking decisions made due to fractional priorities.

Lemma 1. *Integer prioritization, where man m' wins a tie over man m only if $P_i(m') > P_i(m)$, yields a matching M with no augmenting paths of length 3 or shorter.*

Fixing a particular optimal matching M_{OPT} , $M \oplus M_{OPT}$ is the graph formed by taking edges that occur in M or M_{OPT} but not both. An augmenting path is a odd-length path in $M \oplus M_{OPT}$ containing one more edge from M_{OPT} than from M . It is well-known that if all augmenting paths admitted by M have length at least $2k + 1$, then $|M_{OPT}| \leq \frac{k+1}{k}|M|$. This is easy to argue by looking at how much $|M|$ expands when transforming M into M_{OPT} by toggling all the edges in $M \oplus M_{OPT}$: cycles, even-length paths, and odd-length non-augmenting paths in $M \oplus M_{OPT}$ can be toggled without increasing $|M|$, and an augmenting path of length $2n + 1 \geq 2k + 1$ has n edges from M and $n + 1$ from M_{OPT} , so its cardinality expands by a factor of $\frac{n+1}{n} \leq \frac{k+1}{k}$.

Proof of Lemma 1. The GS algorithm itself prevents augmenting 1-paths, since it produces a maximal matching. Any augmenting 3-path must have the structure shown in Figure 1(a) in order for its edges in both M and M_{OPT} to be non-blocking. Woman w_1 clearly never saw a proposal, since any woman receiving a proposal will become and henceforth remain matched. Thus, $P_i(m_1) = 0$. However, since m_2 is unassigned, $P_i(m_2) > 0$, and so w_2 would have chosen m_2 over m_1 . $\square$

Integer prioritization is not sufficient to prevent all augmenting paths of length 5 or longer, although it can prevent some types of these paths from arising. Iwama et al. show that an augmenting 5-path in the specific form shown in Figure 1(b) can occur, since no amount of prioritization on the single man m_3 can induce w_3 to accept him instead of m_2 . Figure 1(c) depicts one of several augmenting 5-paths that cannot occur with $T \geq 3$. To see why, let us number the men and women in any path in $M \oplus M_{OPT}$ sequentially as in Figure 1, so that if the path contains a woman unmatched in M , such a woman is labeled w_1 at the beginning of the path. Edges in M_{OPT} are of the form (m_j, w_j) , and edges in M are of the form (m_j, w_{j+1}) .

Definition 1 (High and low priority for men). *Man m_j in a path in $M \oplus M_{OPT}$ has high priority if $P_i(m_j) \geq j$ at termination; otherwise m_j has low priority.*

Definition 2 (High and low priority for women). *Any woman w_j in a path in $M \oplus M_{OPT}$ unmatched in M has low priority; otherwise if matched in M , woman w_j has high priority if and only if m_{j-1} (her partner in M) has high priority.*

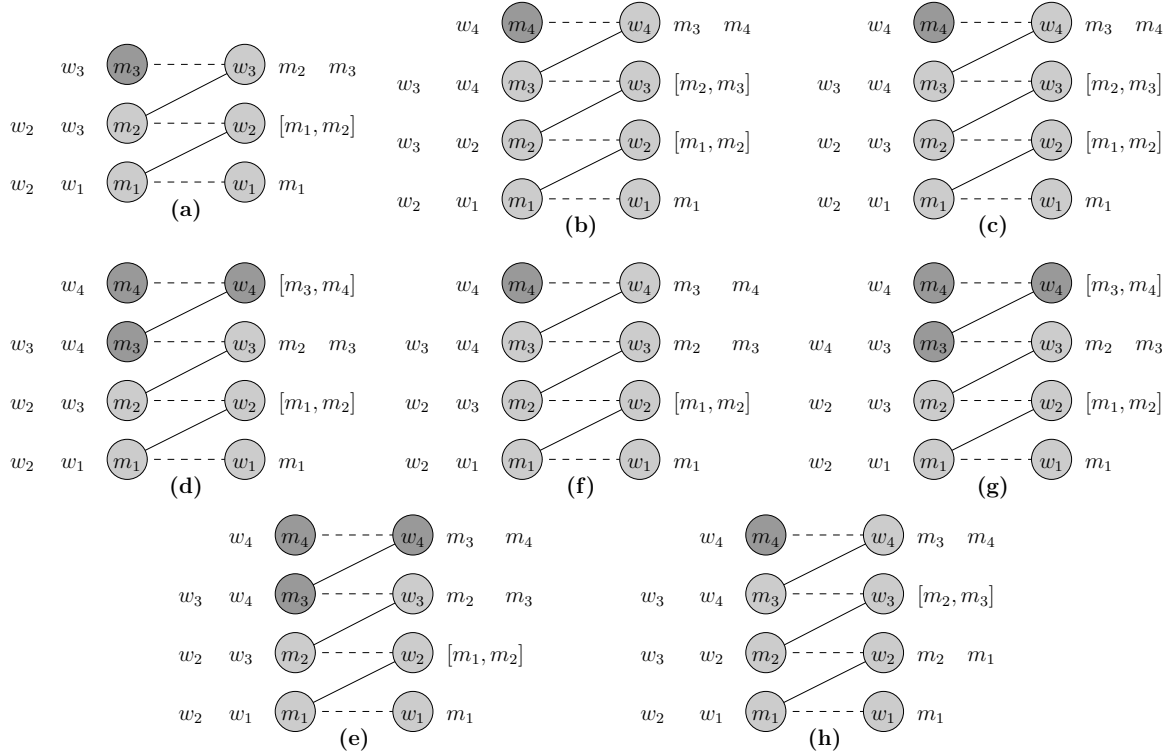


Figure 2: The only possible configurations for (a) an augmenting 5-path (assuming $T \geq 3$), and (b)-(h) augmenting 7-paths (assuming $T \geq 4$). Darkened nodes indicate individuals with high priority.

The following facts are easy to establish:

- (i) $P_i(m_j) = 0$ (and hence m_j has low priority) if he is matched in M_{OPT} to a woman (w_j) unmatched in M . Otherwise, he would have proposed to w_j , and she would have accepted. Since matched women stay matched, w_j could not have ended up single.
- (ii) Man m_j has $P_i(m_j) = 0$ (and hence has low priority) if $m_j \succ_{w_j} m_{j-1}$, for similar reasons. Otherwise, m_j would have proposed to w_j and she would have accepted, preventing the edge (m_{j-1}, w_j) from becoming part of M .
- (iii) If m_j has high priority and $m_j =_{w_j} m_{j-1}$, then m_{j-1} has high priority, since $P_i(m_j) \geq j$ implies that m_j must have been proposed to w_j at least at priority level $j - 1$. Hence, $P_i(m_{j-1}) \geq j - 1$, or else there is no way w_j could have accepted m_{j-1} in the matching M .

If $T \geq 3$, then the highest man (m_3) in an augmenting 5-path will have high priority, since m_3 ends up unmatched, with priority $P_i(m_3) = T$. For The augmenting 5-path in Figure 1(c) therefore cannot occur because by applying (iii) twice, m_1 would have high priority, contradicting (i).

If $T \geq 4$, the highest man (m_4) in an augmenting 7-path will have high priority, and we can similarly deduce that augmenting 7-paths can only occur in the seven configurations shown in Figure 2(b)-(h). One can programatically enumerate all such possible types of augmenting k -paths for larger values of k by filtering out those that would violate the priority rules above, as well as those where edges in M block M_{OPT} or vice versa. These grow at an exponential rate: there are 37 different augmenting 9-paths, 181 augmenting 11-paths, and so on. As we

discover in our analysis, however, we only need to consider up to augmenting 7-paths, so we can set $T = 4$. Raising T higher, while useful for restricting augmenting 9-paths and longer, cannot further help us restrict the set of augmenting 7-paths.

Taking into account the restricted structure of augmenting 5-paths, and by adding fractional prioritization based on the LP relaxation, Iwama et al. were able to show using a fairly complicated “ad hoc” analysis that a solution cannot admit only augmenting 5-paths. Longer paths are also necessary, thereby improving slightly on the $3/2$ approximation ratio one would obtain with only augmenting 5-paths. Our analysis follows a similar pattern, but using a more general approach involving a factor-revealing LP that considers augmenting 7-paths and beyond.

4 Towards a Factor-Revealing LP

In order to analyze the approximation factor of the GS-LP algorithm, we look for an instance I maximizing the ratio $A(I) = |M_{OPT}(I)|/|M(I)|$, where $M(I)$ is the smallest possible matching GS-LP can produce when run on I . We search for I systematically by solving a large LP, known as a factor-revealing LP since its output gives us a valid upper bound on the approximation ratio of the GS-LP algorithm. The factor-revealing LP, which we call FLP, has two types of decision variables, “class” variables (α ’s) and “edge bundle” variables (β ’s).

4.1 Class variables

Fixing an instance I , consider overlaying the edges of $M = M(I)$ and $M_{OPT} = M_{OPT}(I)$. We find a number of different *classes* of structures, shown in Figure 3, which allow us to characterize the edges in M as well as the individuals who are unmatched in M . We find: (a) edges $(m, w) \in M \cap M_{OPT}$, (b) individuals that are unmatched in both M and M_{OPT} , (c) individuals matched in M_{OPT} but not M (these are the endpoints of paths in $M \oplus M_{OPT}$), (d) edges $(m, w) \in M$ where both m and w are matched, differently, in M_{OPT} , and where (m, w) belongs to a path in $M \oplus M_{OPT}$, (e) edges $(m, w) \in M$ where only w is matched in M_{OPT} (these are found at the ends of paths in $M \oplus M_{OPT}$ that are not augmenting paths), (f) edges $(m, w) \in M$ where only m is matched in M_{OPT} (these are found at the ends of paths in $M \oplus M_{OPT}$ that are not augmenting paths), and (g) edges $(m, w) \in M$ found in cycles in $M \oplus M_{OPT}$.

Every edge $(m, w) \in M$ can be categorized into one of these classes, and similarly every man $m \in L$ and woman $w \in R$ in our instance belongs to exactly one of these classes. Observe that we have further differentiated our classes based on the structure of their preference lists with respect to placement of partners in M and M_{OPT} , and also whether individuals are high or low priority (note that this designation is only defined for individuals on paths in $M \oplus M_{OPT}$). For example, the first class shown in Figure 3(f) represents all edges $(m, w) \in M$ in our instance where m is matched to a better partner (w_o) in M_{OPT} than in M , where w is unmatched in M_{OPT} , and where both m and w have low priority.

Let C denote the set of the classes above (note that soon we will enlarge C to contain even more classes, namely explicit representations of all the valid configurations of augmenting 5-paths, 7-paths, etc.). For each class $c \in C$, let g_c denote the number of edges in M represented by a single instance of c . We have $g_c = 1$ for all classes above except those of types (b) and (c), where $g_c = 0$. Let n_c denote the number of occurrences of class c when we overlay M and M_{OPT} , and let the decision variable α_c represent $n_c/|M|$. Since all edges in M appear in exactly one occurrences of some class,

$$\sum_{c \in C} g_c n_c = |M|,$$

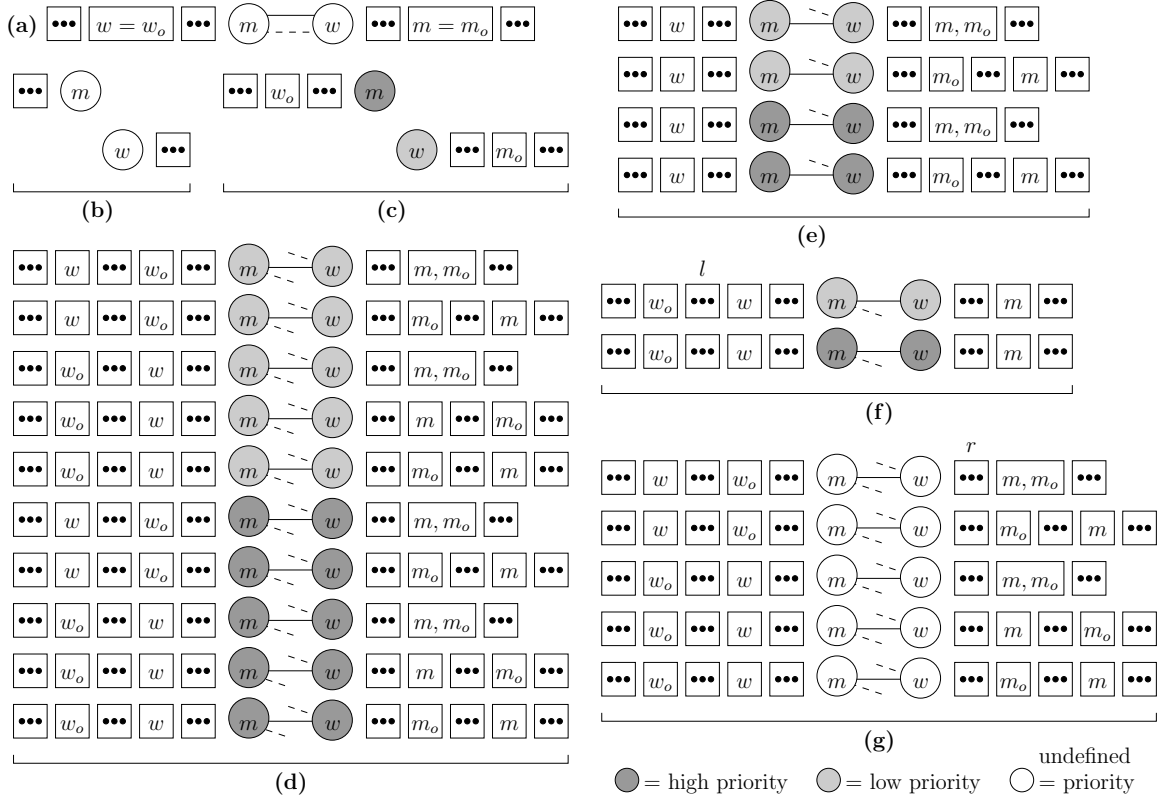


Figure 3: Classes of structures over which the factor-revealing LP optimizes, where m_o and w_o indicate partners in M_{OPT} . Boxes denote distinct “regions” in preference lists, where “...” indicates a region with possibly several entries. Since women can have ties in their lists, a right-hand side region with a specific label like “ m_2, m_3 ” represents a block of preference list entries that are all tied with m_2 and m_3 . In contrast, a left-hand side region with a specific label like “ w_1 ” represents only a single specific entry in a preference list. The pair (l, r) is one example of a pair of regions between which edges in our instance cannot run, since they would block M . The single man in (c) will have high priority provided that T is sufficiently large.

and by dividing both sides by $|M|$, we obtain

$$\sum_{c \in C} g_c \alpha_c = 1, \quad (4)$$

giving a simplex-like constraint in FLP that governs the relative proportion α_c of each class $c \in C$.

Let f_c denote the “local” approximation factor for class c – that is, an upper bound on the factor by which the edges of M belonging to instances of c would expand if we were to transform M into M_{OPT} by toggling the edges in $M \oplus M_{OPT}$. We set $f_c = 0$ by convention for the special cases of (b) and (c), since these involve no edges from M . We have $f_c = 1$ trivially for classes c of type (a), and we also have $f_c = 1$ for classes c of types (e), (f), and (g) since these can only belong to non-augmenting paths or cycles in $M \oplus M_{OPT}$. For classes c of type (d), we have $f_c = 3/2$, since these represent edges in M that might belong to augmenting paths of length 5 or longer (these could also belong to non-augmenting paths, of course, but in this case the local approximation ratio is at most $1 \leq 3/2$). We let C_R denote the 10 classes of type (d), singling them out since they are the only classes so far with a non-trivial local approximation

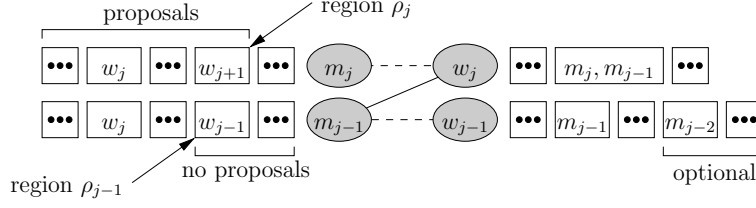


Figure 4: The part of an augmenting path class to which a tie-breaking constraint applies. Woman w_{j-1} could be either unmatched in M or she must prefer her partner m_{j-1} in M_{OPT} to her partner m_{j-2} in M .

ratio. Since every edge in M is represented by an instance of some class $c \in C$, we have

$$|M_{OPT}| \leq \sum_{c \in C} g_c f_c n_c. \quad (5)$$

4.2 Edge Bundle Variables

In addition to the class variables α_c , FLP also contains edge bundle variables β_{lr} that represent the aggregate weight allocated to specific bundles of edges in an optimal solution x^* to $LP(I)$.

In Figure 3, the preference lists in each class $c \in C$ are each partitioned into distinct *regions*. For example, consider again the first of the two classes shown in Figure 3(f), describing an edge $(m, w) \in M$ where m is matched in M_{OPT} (to w_o) but w remains single in M_{OPT} . Here, the preference list of m is divided into five regions: (1) women m prefers more than w_o , (2) the woman w_o , (3) women m prefers less than w_o but more than w , (4) the woman w , and (5) women m prefers less than both w and w_o . The preference list of w is divided into three regions: (1) men she prefers more than m , (2) men she prefers the same amount as m , and (3) men she prefers less than m .

Let Z_L and Z_R respectively denote the sets of all left-hand and right-hand regions across all the classes $c \in C$. For any edge $e = (m, w) \in E$ in our problem instance, we can classify its left endpoint according to its region type $l(e) \in Z_L$ and its right endpoint according to its type $r(e) \in Z_R$. We now define β_{lr} to represent the following quantity:

$$\beta_{lr} = \frac{1}{|M|} \sum_{\substack{e \in E \\ l(e)=l, r(e)=r}} x_e^*. \quad (6)$$

That is, β_{lr} denotes the aggregate weight assigned by $LP(I)$ to the bundle of all edges from region type $l \in Z_L$ to region type $r \in Z_R$. Note that not every pair of regions $(l, r) \in Z_L \times Z_R$ is *valid* in its ability to describe edges from E . The following pairings (l, r) are invalid, since there cannot be any edges $e \in E$ satisfying $l = l(e)$ and $r = r(e)$: (i) region pairs (l, r) corresponding to edges that would block M (for example, the pair (l, r) shown in Figure 3), (ii) region pairs (l, r) corresponding to edges that would block M_{OPT} , (iii) region pairs (l, r) corresponding to edges from a high-priority man m to a woman w , where m is preferred to her partner in M (this is not possible since $P_i(m) > 0$, so m would have proposed to w , and w would have accepted, preventing later acceptance of her partner in M), (iv) region pairs (l, r) corresponding to edges from a high-priority man m to a low-priority woman w , where m is tied with her partner in M (similarly, m would have proposed to w , and w would have accepted, preferring m to her partner in M), and (v) region pairs (l, r) where l is a region representing a single specific individual

(e.g., one's partner w_o in M_{OPT}), but r is not a compatible match for that individual (e.g., not labeled with w_o).

Let $B \subset Z_L \times Z_R$ denote the set of all valid region pairs that remain after filtering out the invalid pairs enumerated above. Since every edge $e \in E$ appears in exactly one such pair $(l(e), r(e)) \in B$, we have

$$\sum_{(l,r) \in B} \beta_{lr} = \frac{1}{|M|} \sum_{e \in E} x_e^*. \quad (7)$$

4.3 The Factor-Revealing LP

We can now state FLP:

$$FLP = \max \left\{ \min \left(\sum_{c \in C} g_c f_c \alpha_c, \sum_{(l,r) \in B} \beta_{lr} \right) : (\alpha, \beta) \in P \right\}, \quad (8)$$

where P denotes its feasible region, described with 5 types of constraints:

- *Nonnegativity* constraints: $\alpha, \beta \geq 0$.
- A *simplex* constraint: $\sum g_c \alpha_c = 1$.
- *Tie-breaking* constraints, stating that weights assigned to edge bundles would have resulted in fractional priorities leading to tie-breaking that would have resulted in the formation of augmenting paths. Consider any part of an augmenting path class matching the situation described by Figure 4. Note that $P_i(m_{j-1}) = 0$, since m_{j-1} must never have proposed to w_{j-1} . Therefore, the fact that w_j chose m_{j-1} over m_j implies that $P_i(m_j) = 0$ and so $P'_f(m_j) \leq P'_f(m_{j-1})$ where P'_f denotes P_f at the last point in time at which the algorithm broke the tie. Since m_j is matched with w_{j+1} in M , his fractional priority at that point in time was equal to the weight of all entries in his preference list prior to w_{j+1} :

$$P'_f(m_j) = \sum_{\substack{e=(m_j, w) \in E \\ l(e) \succ \rho_j}} x_e^*,$$

where ρ_j is the region of m_j 's list containing w_{j+1} , as indicated in the figure. Since m_{j-1} never proposed to w_{j-1} and also never proposed to any woman w matched in M to a man m for which $m_{j-1} \succ_w m$, we have

$$P'_f(m_{j-1}) \leq \sum_{\substack{e=(m_{j-1}, w) \in E \\ l(e) \succ \rho_{j-1}, r(e) \notin Z_R^M}} x_e^*,$$

where $Z_R^M \subset Z_R$ denotes the set of all right-hand side regions that are more preferred than the partner in M within their same list. Combining these, we have

$$\sum_{\substack{e=(m_j, w) \in E \\ l(e) \succ \rho_j}} x_e^* \leq \sum_{\substack{e=(m_{j-1}, w) \in E \\ l(e) \succ \rho_{j-1}, r(e) \notin Z_R^M}} x_e^*,$$

which in aggregate across all class instances becomes the following tie-breaking constraint in FLP:

$$\sum_{\substack{(l,r) \in B \\ l \succ \rho_j}} \beta_{lr} \leq \sum_{\substack{(l,r) \in B \\ l \succ \rho_{j-1}, r \notin Z_R^M}} \beta_{lr}.$$

- *Matching* constraints, stating that the total weight incident to any generic man or woman in class c must not exceed α_c . Consider some generic man m belonging to class c , and let $L(m)$ denote the set of n_c individual men in L represented by the generic man m . Since each individual man $m' \in L(m)$ satisfies constraint (1) in $LP(I)$, we sum these constraints to obtain

$$\sum_{\substack{(m',w) \in E \\ m' \in L(m)}} x_{m'w}^* \leq n_c,$$

Therefore,

$$\sum_{\substack{(l,r) \in B \\ l \in Z_L(m)}} \beta_{lr} = \frac{1}{|M(I)|} \sum_{\substack{(m',w) \in E \\ m' \in L(m)}} x_{m'w}^* \leq \frac{n_c}{|M(I)|} = \alpha_c,$$

where $Z_L(m)$ denotes the regions in m 's preference list. Similar constraints hold for the women.

- *Stability* constraints, which correspond to aggregations of the stability constraints from the original SMTI LP relaxation across our edge bundle variables. Consider any class c and region pair (l, r) representing an edge explicitly found within c . So far, this includes just the single edge $(m, w) \in M$ represented by class c , but soon we will expand C by including larger augmenting structures that also explicitly include edges from M_{OPT} . We must have

$$\sum_{\substack{(l',r') \in B \\ l' \succ l}} \beta_{l'r'} + \sum_{\substack{(l',r') \in B \\ r' \succeq r}} \beta_{l'r'} \geq \alpha_c, \quad (9)$$

where the notation $r' \succ r$ indicates that r' is a region more preferred than r belonging to the same preference list. That is, the total weight incident to regions in the preference list of l preceding l , plus the total weight incident to regions in the preference list of r up to and including r , must be at least α_c .

5 Analysis and Refinement of FLP

Lemma 2. *For any instance I , let α and β be defined as above. Then $(\alpha, \beta) \in P$.*

Proof. Nonnegativity of α and β is clear, validity of the simplex constraint is shown in (4), and validity of the tie-breaking and matching constraints are argued above. Validity of the stability constraints are shown exactly the same way, by aggregating constraint (3) from $LP(I)$ over all n_c individual instances of an edge appearing explicitly within some class c . $\square$

We now establish our main result:

Theorem 3. *Let $A(I)$ denote the approximation ratio of GS-LP when run on instance I . Then $FLP \geq A(I)$.*

Proof. Let α, β be defined as above. Lemma 2 shows that $(\alpha, \beta) \in P$. We now have

$$\sum_{c \in C} g_c f_c \alpha_c = \frac{1}{|M(I)|} \sum_{c \in C} g_c f_c n_c \geq \frac{1}{|M(I)|} |M_{OPT}(I)| = A(I),$$

where the first equality follows from the definition of α_c , and the next inequality is a re-statement of (5). From (7), we also have

$$\sum_{(l,r) \in B} \beta_{lr} = \frac{1}{|M(I)|} \sum_{e \in E} x_e^* = \frac{1}{|M(I)|} LP(I) \geq \frac{1}{|M(I)|} |M_{OPT}(I)| = A(I).$$

Hence,

$$\min \left(\sum_{c \in C} g_c f_c \alpha_c, \sum_{(l,r) \in B} \beta_{lr} \right) \geq A(I),$$

so we have demonstrated a feasible solution of FLP with objective value at least $A(I)$. The maximum value of FLP is therefore also an upper bound for $A(I)$. $\square$

5.1 Adding More Classes

The optimal solution value of FLP as stated above is $3/2$, and this is not surprising, since it really isn't capturing any insight beyond the fact that there are no augmenting 3-paths. To improve it, we need to add more explicit knowledge about the structure of augmenting 5-paths, so that it can deduce (like the work of Iwama et al.) that it is not possible to form an instance I admitting only augmenting 5-paths.

We therefore add a new class c representing the unique structure of an augmenting 5-path to C , with $g_c = 2$ and $f_c = 3/2$. Accordingly, we reduce f_c to $4/3$ for all classes $c \in C_R$ – since we are now explicitly representing augmenting 5-paths as their own class, the classes in C_R now represent the “left-over” edges in M that are at worst now part of augmenting 7-paths. Solving FLP now yields an optimal solution of value $25/17$, matching the result of Iwama et al. In fact, we can think of FLP in this case as automating the proof of Iwama et al. Instead of starting with known inequalities (e.g., the constraints from the LP relaxation) and combining them in an “ad hoc” series of lemmas and theorems, FLP automatically finds the best combination.

To improve further, we can add more classes to C that explicitly represent all possible arrangements of augmenting 7-paths, reducing f_c to $5/4$ for all $c \in C_R$. This yields an optimal solution of value $19/13 \approx 1.4615$. One might hope that this process would continue reducing the upper bound on our approximation ratio in an asymptotic manner, since each time we explicitly include longer augmenting paths, the optimum solution of FLP cannot increase.

Lemma 4. *Let FLP_k denote FLP when up through augmenting k -paths are explicitly represented. Then $FLP_{k+2} \leq FLP_k$.*

Proof. This follows from the fact that FLP_k is a relaxation of FLP_{k+2} . That is, the optimal solution (α, β) for FLP_{k+2} can be naturally mapped to a feasible solution (α', β') for FLP_k (by assigning all weight from classes representing augmenting $(k+2)$ -paths to the corresponding classes in C_R), for which $FLP_k \geq FLP_k(\alpha', \beta') \geq FLP_{k+2}(\alpha, \beta) = FLP_{k+2}$. $\square$

Unfortunately, when we solve FLP after explicit inclusion of augmenting 9-paths and beyond, we still obtain an optimal solution of $19/13$, and indeed this is the best result we can obtain, since when we build FLP up to just augmenting 7-paths, we find an optimal solution in which $\alpha_c = 0$ for all $c \in C_R$.

Lemma 5. *If an optimal solution of FLP_k assigns $\alpha_c = 0$ for all $c \in C_R$, then $FLP_k = FLP_{k'}$ for all $k' \geq k$.*

Proof. The fact that $FLP_k \geq FLP_{k'}$ comes from Lemma 4. To show the reverse inequality, observe that any optimal solution (α, β) for FLP_k with $\alpha_c = 0$ for all $c \in C_R$ can be naturally mapped to an equivalent solution (α', β') of $FLP_{k'}$ with $FLP_k = FLP_k(\alpha, \beta) = FLP_{k'}(\alpha', \beta') \leq FLP_{k'}$, since only the classes $c \in C_R$ change their meaning and local approximation ratio f_c when moving from k to k' . $\square$

To ensure that our results are accurate (due to the highly complex nature of FLP), we built three independent code bases for generating and solving FLP, all of which agreed in their output.

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